

HW/SW Codesign

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- Seminar
 - Based on a given topic
 - Paper (5 pages) plus seminar presentation
 - IEEE latex template
(<https://www.ieee.org/conferences/publishing/templates.html>)
 - Presentation 10 Min + 5 Min discussion
 - Paper written in scientific way
 - Prerequisite: passed scientific work module
 - We will not repeat details on this.
 - Plagiarism: illegal -> failed, triggers internal process, wc ex-matriculated!
 - i.e. every sentence of the paper is of intellectual origin of the author and not a form of copy (direct or intellectual) from any other author (no matter from which resource).
 - Use citavi or Zotero for managing your references
 - Write for each related reference a short note of how it is related to your seminar topic – the note is part of your bibtex management tool and/or an extra bibtex entry.
 - Use github! (achim.rettberg@iess.org)
 - Rest of semester: preparation of paper/topic and presentation plus lab work
 - Details follows

► Possible Topics

- List-Based Scheduling for High-Level Synthesis
- ILP-Based Scheduling for High-Level Synthesis
- Force-Directed Scheduling for High-Level Synthesis
- Force-Directed List-Scheduling for High-Level Synthesis
- Multiprocessor Scheduling & Hu's Algorithm for High-Level Synthesis
- High-Level Synthesis for Scheduling Pipelined Circuits
- High-Level Synthesis with Simulated Annealing Scheduling
- High-Level Synthesis with Genetic Algorithm Scheduling
- High-Level Synthesis with Geometric Algorithm Scheduling
- High-Level Synthesis with Game Theoretic Scheduling
- High-Level Synthesis with Tabu Search Scheduling
- High-Level Synthesis with Ant Colony Optimization Scheduling
- Sharing and Binding within High-Level Synthesis
- Low Power Scheduling for High-Level Synthesis
- Loop Unrolling in High-Level Synthesis
- Dataflow-Optimization within High-Level Synthesis
- Memory Optimization within High-Level Synthesis
- High-Level Synthesis for Multi-Core Systems
- High-Level Synthesis for FPGA
- High-Level Synthesis for Edge-AI Applications
- Hardware Acceleration and Sensor Fusion with High-Level Synthesis

- Familiarize yourself with the topic based on the main reference.
- Check additional references
 - Books, paper, ... (see hshl bib, oreilly, ieee, acm, ...)
- The elaboration consists of the following parts
 - Necessary mathematical basics
 - Formal (mathematical) description of the topic
 - Formal description of the algorithm
 - Runtime / overhead of the algorithm
 - Implementation of main feature in C/C++
 - Realization of an application example
 - Output of High-Level Synthesis is a netlist of Register-Transfer Level (RTL) components (an RTL component can be a VHDL entity)